PROOF OF CONCEPT · MALL SECURITY CAMERAS · OPUS 5.5 VISION · 21 OCTOBER 2026

Spill detection on the mall's own cameras

The mall already has fixed cameras over every corridor and atrium. This proof of concept uses them to spot a liquid spill within seconds of it happening and to send the cleaning robot only to confirmed spills, so the robot does not patrol. No training data was collected and no model was trained: a general vision model judges what changed on the floor against a written description of a spill, and every answer is kept as a label for a smaller on-site model later.

98%
accurate on mounted cameras
305 before-and-after checks, 4 cameras

96%
of spills found
131 of 136 rendered spills on real floors

0
false alarms
in 169 real frames of people and trolleys

$0.05
per checked frame
frames with nothing persistent cost nothing

HOW IT WORKS

Reference, persistent change, verdict

Because the cameras never move, the server keeps a reference picture of each empty floor and only asks the model about patches that changed and *stayed* changed: a spill persists, a person walks on. Each such patch is cut out of the frames at full size and sent with the reference and a frame two seconds later. A frame with no persistent change costs nothing. The robot is dispatched only to a confirmed spill.

1 · Reference

Remember the empty floor

A median over a few seconds of footage removes passers-by and leaves the clean floor, per camera. Refreshed slowly so lighting changes do not pile up.

2 · Persistent change

Find what appeared and stayed

Difference against the reference, kept only where the change is still there two seconds later. People, trolleys and their shadows drop out; spills, dropped bags and signs remain.

3 · Ask about the patch

Three frames, one crop

Claude sees the same crop from the reference, the current frame and a frame two seconds later, and answers "has a liquid spill appeared on the floor?" with the same JSON as before. The third frame is how it tells a spill, which stays, from a shadow, which walks away. Boxes come back in frame coordinates.

RESULTS

Across all 458 pairs from 5 cameras: **97%** accuracy, 205 of 213 spills found, 8 false alarms in 245 real frames, and 200 of 205 found spills placed at the right spot. On the four mounted cameras alone (305 pairs): **98%** accuracy, 131 of 136 spills found, 0 false alarms in 169 real frames. The handheld walkway clip accounts for every false alarm.

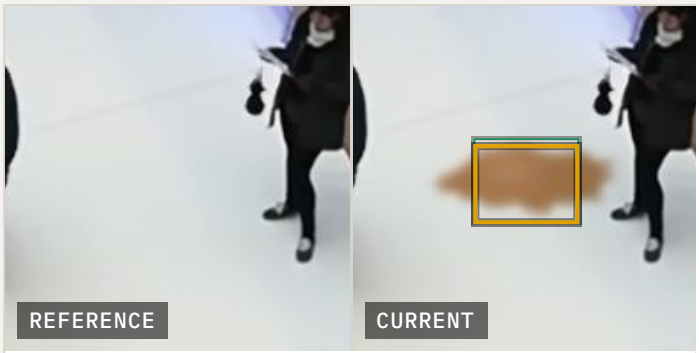
Pairs below are exactly what the model saw: the same crop from the reference, the current frame and two seconds later. The amber box is the model's answer, the thin green box is the true spill. Captions name the camera and the truth.



SPILL confidence 0.93 Found

A large tan/brown puddle with an irregular outline has appeared on open floor below the escalator. It is absent from the reference and unchanged in the later frame, and no person touches it, so it is not a shadow.

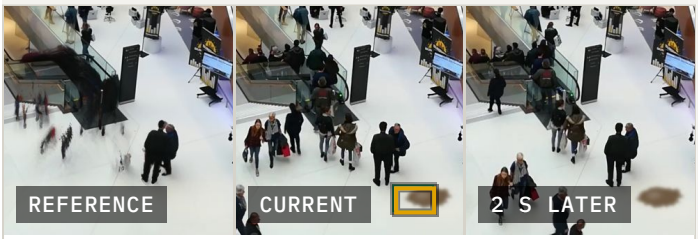
atrium_escalators camera · truth: synthetic juice spill, 11% of frame width



SPILL confidence 0.93 Found

A brownish irregular puddle has appeared on open floor between the people. It is absent in the reference, unchanged in the later frame, and does not touch anyone's feet, so it is not a shadow.

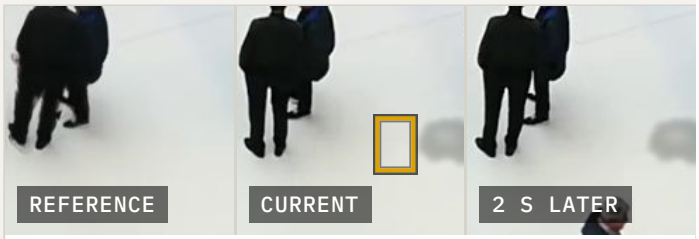
atrium_escalators camera · truth: synthetic coffee spill, 5% of frame width



SPILL confidence 0.90 Found

A brown, irregular puddle-shaped patch is on open floor at the lower right in the current frame. It is absent from the reference, unchanged in the later frame, and not attached to any person, which fits a spilled drink such as coffee.

atrium_escalators camera · truth: synthetic cola spill, 6% of frame width



SPILL confidence 0.88 Found

A grey, irregular, puddle-shaped dark patch has appeared on open floor right of centre. It is absent in the reference, is not attached to any person, and stays unchanged in the later frame, which is consistent with a liquid spill.

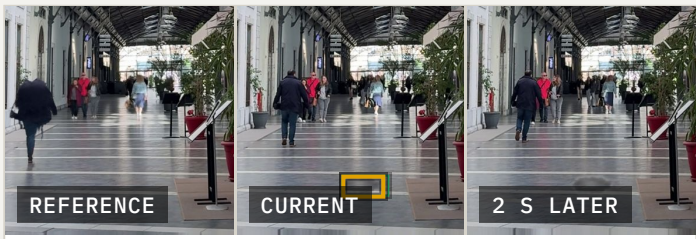
atrium_escalators camera · truth: synthetic water spill, 4% of frame width



SPILL confidence 0.90 Found

A new yellowish-brown patch with an irregular rounded outline appears on open floor in the current frame and is unchanged in the later frame. No person is nearby to cast it, so it looks like a spilled drink such as juice or tea.

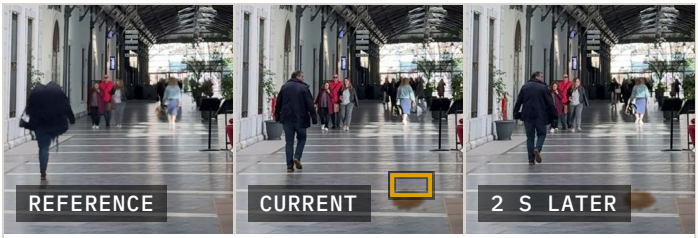
corridor_vert camera · truth: synthetic juice spill, 9% of frame width



SPILL confidence 0.85 Found

A new dark, irregular, wet-looking patch has appeared on open floor near center-bottom that is absent in the reference. It persists unchanged in the later frame and is not attached to any person, so it is not a moving shadow.

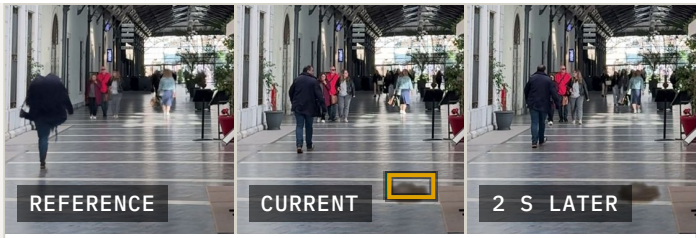
corridor_vert camera · truth: synthetic water spill, 10% of frame width



SPILL confidence 0.85 Found

A brown, irregular patch with a liquid-like outline has appeared on open floor at the lower right. It is absent from the reference and unchanged in the later frame, and no person is near it, so it is not a shadow.

corridor_vert camera · truth: synthetic coffee spill, 8% of frame width



SPILL confidence 0.85 Found

A new brownish patch on open floor near the right mat is absent in the reference and unchanged in the later frame, with no person touching it. Its dark, wet-looking outline suggests a liquid or semi-liquid spill.

corridor_vert camera · truth: synthetic cola spill, 10% of frame width



SPILL confidence 0.93 Found

A brownish, irregular puddle has appeared on open floor in the lower left that is absent from the reference. It stays unchanged in the later frame and is not attached to any person, which fits a spilled drink such as coffee or juice.

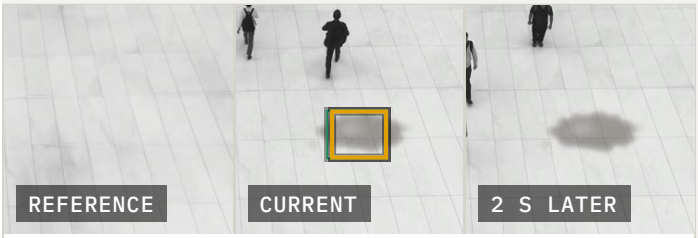
plaza_overhead camera · truth: synthetic coffee spill, 7% of frame width



SPILL confidence 0.93 Found

A yellow-orange irregular patch with a splash-like outline is on open floor in the current frame. It is absent from the clean reference, does not touch either person, and is still present unchanged after both people have left, so it is a liquid spill such as juice.

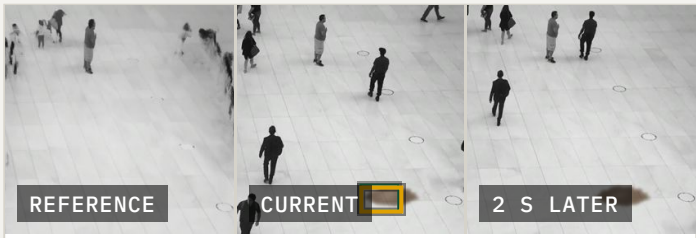
plaza_overhead camera · truth: synthetic juice spill, 6% of frame width



SPILL confidence 0.90 Found

A dark, irregular puddle-shaped patch on open floor is absent from the reference and appears unchanged in both later frames, with no person touching it. That rules out a shadow or a moving object.

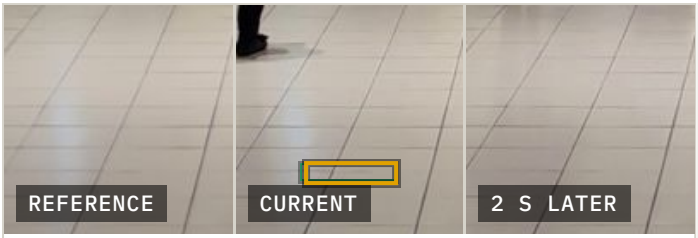
plaza_overhead camera · truth: synthetic oil spill, 11% of frame width



SPILL confidence 0.90 Found

A brown, irregular puddle-shaped patch on open floor near the bottom centre is absent in the reference but present and unchanged in both the current and later frames, and it does not touch any person. Its brown colour suggests coffee or a similar drink.

plaza_overhead camera · truth: synthetic cola spill, 7% of frame width



SPILL confidence 0.93 Found

A brownish puddle with an irregular splash outline appears on open floor in the current frame. It is absent from the reference and unchanged in the later frame, it is not connected to any person, and it looks like a spilled drink such as coffee.

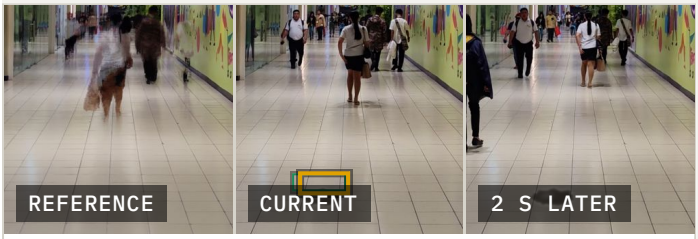
walkway_vert camera · truth: synthetic coffee spill, 6% of frame width



SPILL confidence 0.93 Found

A new amber/orange puddle with an irregular splash outline sits on open tile floor, absent from the reference and unchanged in the later frame. Its colour suggests juice or a soft drink, and no person or object is nearby to cast a shadow.

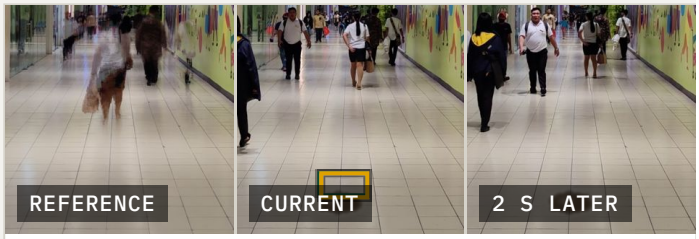
walkway_vert camera · truth: synthetic juice spill, 11% of frame width



SPILL confidence 0.90 Found

A dark, irregular, wet-looking patch on open floor (lower centre) is absent from the reference and identical in the current and later frames, while the people around it have moved. It has its own outline and touches no person, so it is not a shadow or a wall reflection.

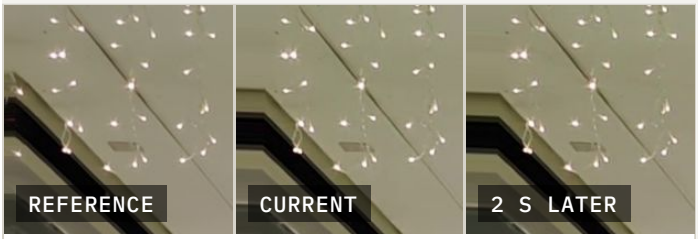
walkway_vert camera · truth: synthetic oil spill, 9% of frame width



SPILL confidence 0.90 Found

A new brownish puddle with an irregular, darker rim appears on open floor in the current frame, absent from the reference. It is unchanged in the later frame and has no person nearby, so it is not a shadow.

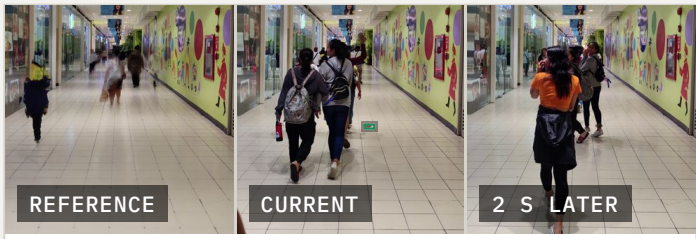
walkway_vert camera · truth: synthetic cola spill, 10% of frame width



CLEAR confidence 0.96 Missed

All three frames show a ceiling with hanging string lights and an upper shopfront; no floor is visible, so no spill can be assessed. Differences between frames are minor camera shifts only.

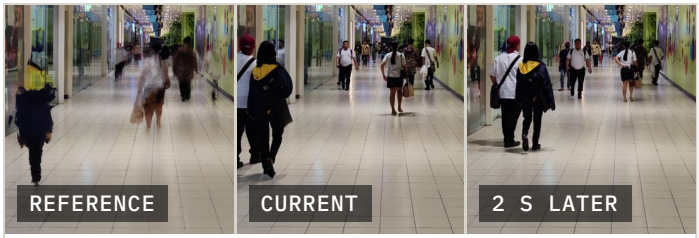
walkway_vert camera, a miss · truth: synthetic oil spill, 4% of frame width



CLEAR confidence 0.70 Missed

The current frame shows a small brownish patch on open floor (around x≈0.57, y≈0.52), but it is absent in the later frame at the same location, so it is not a persistent spill. The other dark patches are shadows of the people walking.

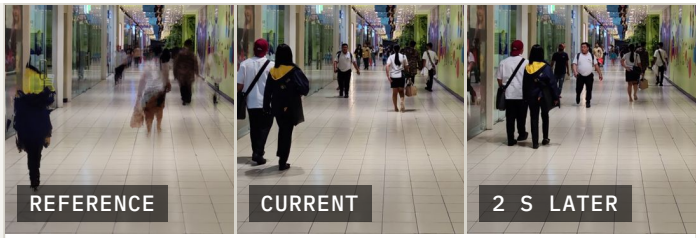
walkway_vert camera, a miss · truth: synthetic cola spill, 3% of frame width



CLEAR confidence 0.93 Correctly clear

The only new dark patches are shadows at the walking people's feet, and they move with them in the later frame. The open tile floor otherwise matches the reference, with only glossy reflections along the shopfronts and wall.

walkway_vert camera, people walking through · truth: no spill, real footage



CLEAR confidence 0.93 Correctly clear

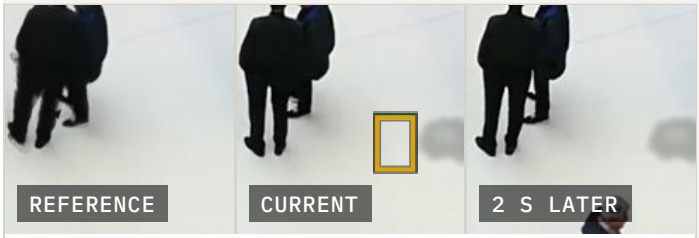
The only new dark patches in the current frame are shadows attached to the walking people's feet, and they move with them in the later frame. The remaining changes are normal reflections on the polished tiles, which are also present in the reference, and no independent wet patch appears.

walkway_vert camera, people walking through · truth: no spill, real footage

CLEAR LIQUIDS

Water is the hard case, and it is found

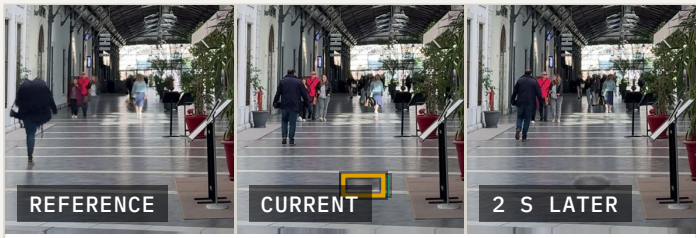
Water has no colour of its own. On a floor it shows only as a darkening of the surface, a soft outline and a slight sheen, which is also how a shadow or a polished patch looks. The model found **50 of 53** water spills across the mall cameras, down to one 3% of the frame wide. Its reasoning on these consistently names the darkening and the fact that the patch stays put and touches no one. Coloured liquids for comparison: coffee 36 of 37, cola 33 of 35, juice 41 of 41, oil 45 of 47.



SPILL confidence 0.88 Found

A grey, irregular, puddle-shaped dark patch has appeared on open floor right of centre. It is absent in the reference, is not attached to any person, and stays unchanged in the later frame, which is consistent with a liquid spill.

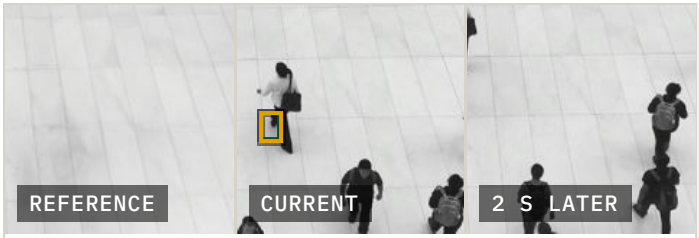
atrium_escalators camera, water · truth: synthetic water spill, 4% of frame width



SPILL confidence 0.85 Found

A new dark, irregular, wet-looking patch has appeared on open floor near center-bottom that is absent in the reference. It persists unchanged in the later frame and is not attached to any person, so it is not a moving shadow.

corridor_vert camera, water · truth: synthetic water spill, 10% of frame width



SPILL confidence 0.88 Found

A dark, irregular puddle-shaped patch on open floor at left-centre is absent in the reference and unchanged in both later frames, with no person nearby, consistent with a liquid spill.

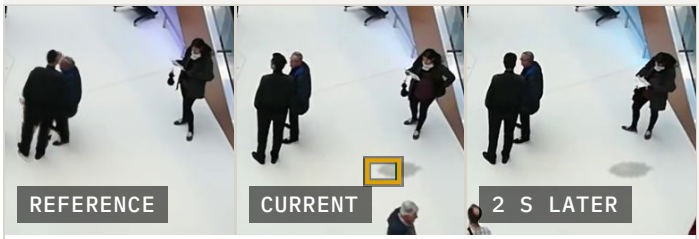
plaza_overhead camera, water · truth: synthetic water spill, 4% of frame width



SPILL confidence 0.90 Found

A dark, irregular, wet-looking patch has appeared on open floor to the right of the walker. It is absent in the reference and unchanged in the later frame after the person moved away, so it is not a shadow.

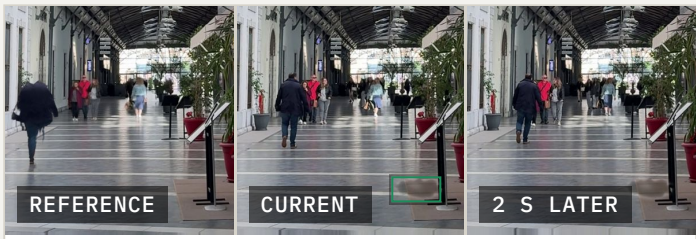
walkway_vert camera, water · truth: synthetic water spill, 6% of frame width



SPILL confidence 0.88 Found

A new grey, irregular-edged patch has appeared on open white floor and is absent from the reference. It is not touching any person and stays in the same place and shape in the later frame, consistent with a liquid puddle.

atrium_escalators camera, the smallest water spill found, 3% of frame width · truth: synthetic water spill, 3% of frame width



CLEAR confidence 0.90 Missed

Across all three frames the polished floor shows only the same light reflections, white stripes and moving pedestrians with their reflections; no new patch with its own liquid outline appears on open floor. A potted plant near the left wall appears in later frames but is not a spill.

corridor_vert camera, a water spill it missed · truth: synthetic water spill, 11% of frame width

	THE PIPELINE: PERSISTENT CROPS, THREE FRAMES (458 PAIRS)	WHOLE FRAME, TWO FRAMES, NO CANDIDATE STEP (153 PAIRS, EVERY THIRD)
Spills found	**205 of 213**	55 of 70
False alarms	**8 of 245**	0 of 83
Spills smaller than 7% of frame width found	**87 of 90**	19 of 27
Model calls per frame	6.1	1
Cost per checked frame	$0.051	$0.016
Latency per frame	21.5 s	10.0 s

APPENDIX

Every check on the mall cameras

All 458 checks, spills first, then the ones the model got wrong, then by camera. Each shows the crop the model judged in the current frame: thin green is the true spill, amber is the model's answer. Hover for its reason.

COST AT SCALE

What this costs across a mall

The camera server does the cheap work for free: it keeps the reference, finds changes and drops the ones that walk away. Only persistent patches reach the model, and each check costs about $0.051 at the 6.1 crops per check seen in the busy test clips. One camera, a 12-hour day:

ONE CAMERA	CHECKS PER DAY	PER DAY	PER MONTH
Quiet floor: one check every 4 minutes	180	$9.25	$277
Busy corridor: one check every minute	720	$37.00	$1,110

The per-check cost was measured on crowded test clips where people standing still add crops; an empty floor costs less per check. Once enough labelled patches have accumulated, a small on-site model takes the easy decisions and the cloud model becomes a second opinion.

NEXT

Roadmap

Done	**Fixed-camera pipeline.** Reference, persistent change, before-and-after crops. Tested on five mall cameras with rendered spills: no false alarms on the mounted cameras.
Next	**One real mall camera, real spills.** A few hours of footage from a single camera in any mall, with water, coffee and oil spilled on the floor in view. This replaces the rendered positives and gives the true recall number and the real check rate.
Then	**Hard negatives at scale.** Polished stone, reflections, cleaning carts, wet-floor signs, dropped bags. Measure the false-alarm rate per camera per day, which is what decides whether staff trust the alerts.
Later	**Camera-server deployment.** Reference and persistence run on the camera server at no cost; only persistent patches go to the model. Robot dispatch on confirmed spills. An on-site model trained from the accumulated labels where the cloud call is not wanted.

ALSO TESTED

A close camera on the robot itself

Before the camera question was settled, the same model was tested on single close-range frames, the view a camera on the robot would have. It is kept here because it is the fallback if a mall's cameras cannot be used, and because it shows how the model behaves on a public spill dataset with real liquids.

CLOSE CAMERA · PIPELINE

How a single frame becomes a verdict

1 ·Sample

Pull frames from the camera

One or two frames per second is enough for a slow-moving robot. Near-identical frames are dropped so a parked robot does not keep paying to look at the same tile.

2 ·Ask

Vision model reads the frame

Claude Opus 5.5 receives the image and a written brief: what counts as a spill, what does not (reflections, shadows, dark tiles, wet-floor signs, liquid in cups). It must answer in a fixed JSON shape.

3 ·Act and keep

Verdict, box, reasoning

The robot gets a yes/no, a confidence, a liquid type and a box. The answer is also written to disk with the frame, so every run grows the training set for a small on-board model later.

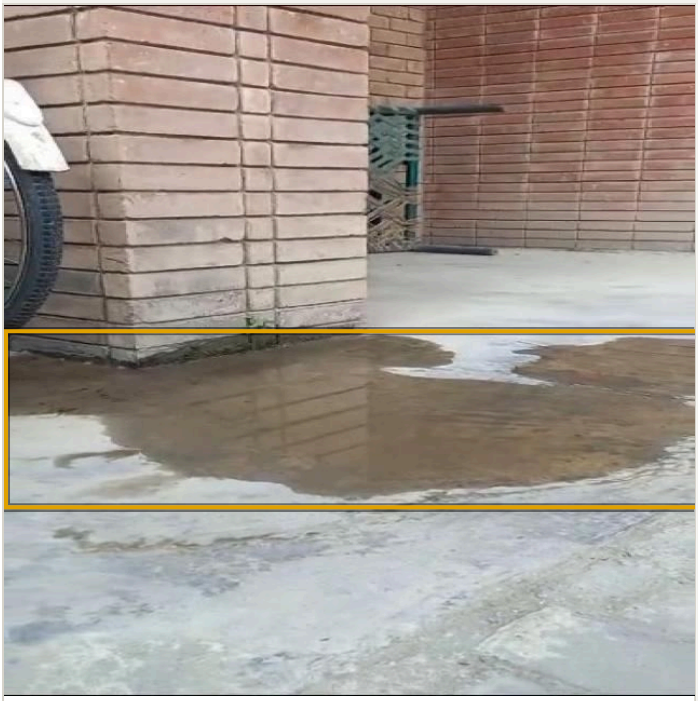
What one answer looks like, for the first frame in the gallery below:


```
{
  "spill_detected": true,
  "confidence": 0.95,
  "spill_type": "water",
  "size": "large",
  "bbox": [
    0.0,
    0.47,
    1.0,
    0.25
  ],
  "debris_present": false,
  "floor_visible": true,
  "reasoning": "A large, irregularly shaped standing puddle of muddy water covers the co
}
```


CLOSE CAMERA · GALLERY

What the model saw, and what it said

Real close-range frames from the public dataset. The amber box is where the model says the spill is. The text under each frame is the model's own one-line reason, unedited.



SPILL confidence 0.95 2.8s

water · large size

A large, irregularly shaped standing puddle of muddy water covers the concrete floor in front of the brick wall, with the wall clearly reflected in its surface. The distinct edges and reflection indicate real pooled liquid rather than a floor pattern or shadow.

Rain puddle on outdoor pavement, phone video at knee height



SPILL confidence 0.93 2.6s

water · large size

A large clear puddle covers the lower half of the concrete floor, with mirror-like reflections of the person's boots and a distinct irregular edge showing standing liquid rather than glossy flooring. The clear colour suggests water.

Person walking through a puddle on a glossy indoor floor

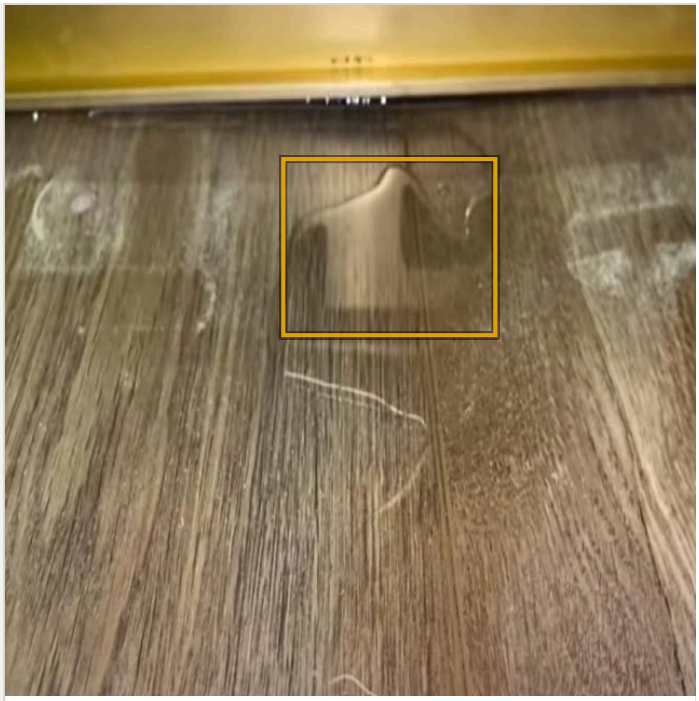


SPILL confidence 0.93 2.7s

water ·large size

A large dark wet patch covers most of the concrete floor, with irregular puddle edges and visible reflections of the motorcycles and plate, which indicates standing water. Dry lighter concrete is visible at the top and bottom-right edges.

Wet concrete outside a parking area, low-angle video



SPILL confidence 0.85 3.9s

water ·medium size ·debris

A clear liquid puddle with a distinct raised meniscus edge and light reflection sits on the wood-look floor near the yellow baseboard. Dried white residue marks and a hair-like strand are also visible.

Small water patch on a wooden floor, screenshot from video



SPILL confidence 0.90 3.3s

water ·large size

A large dark wet splash pattern with radiating streaks and droplet spatter covers most of the concrete floor, consistent with spilled liquid (likely water) rather than a material stain or shadow.

Muddy splash across a tiled floor, WhatsApp video frame



SPILL confidence 0.90 3.2s

water ·medium size

A tipped-over glass lies on a wood-grain surface with a large clear puddle with defined edges and scattered droplets spreading from its rim, consistent with spilled water. The close-up stock-photo framing makes it slightly uncertain whether the surface is a floor rather than a table.

Tipped glass with water spreading on a grey surface



SPILL confidence 0.96 3.9s

water ·medium size

Clear beaded puddles and droplets with raised edges and specular highlights sit on the wood-look plank floor, consistent with spilled water. They are spread across the upper and central frame, with liquid also running along a plank seam.

Water droplets and a wet patch on a dark wooden floor

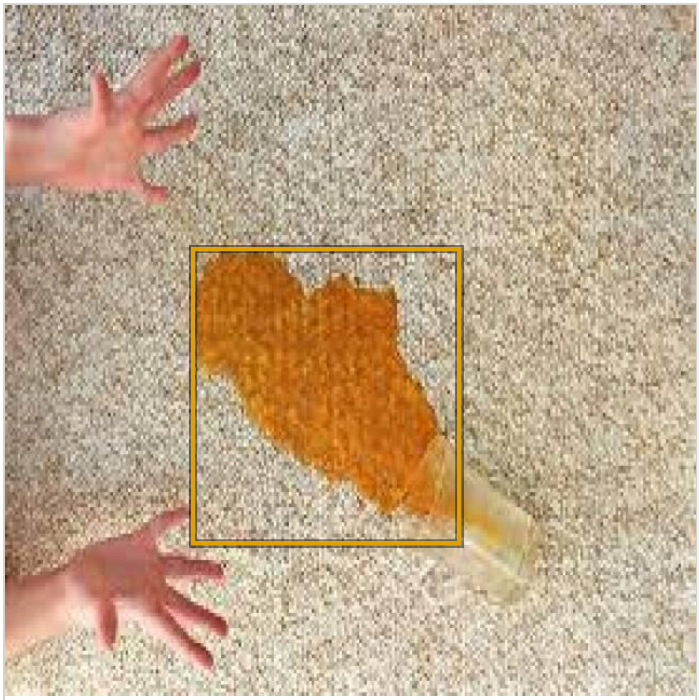


SPILL confidence 0.96 4.3s

other liquid ·medium size

A glossy red liquid pool with many scattered droplets and splatter is visible on light speckled vinyl flooring, consistent with blood or a red drink spill. This may be a biohazard and should be flagged for staff.

Red liquid splashed on white tiles



SPILL confidence 0.96 3.9s

food or drink ·medium size

An orange liquid, likely juice, has soaked into the beige carpet in a distinct stain spreading from a tipped-over glass. The outline and saturation clearly indicate a fresh drink spill.

Orange drink soaked into carpet, hands reaching in

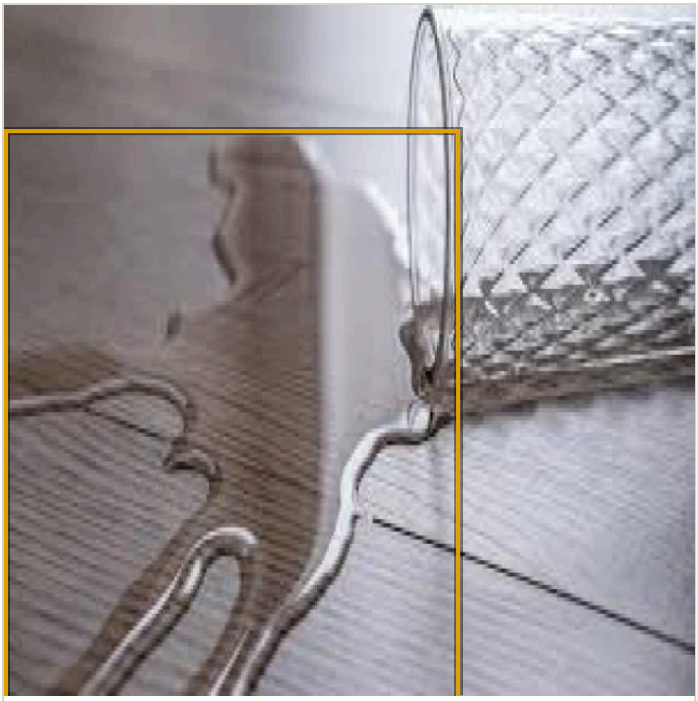


SPILL confidence 0.97 3.9s

food or drink ·large size ·debris

A large brown puddle with irregular edges and surface bubbles spreads across a light floor next to a tipped-over blue mug, consistent with spilled coffee or tea. The overturned mug is an object on the floor that also needs picking up.

Juice pooled on a tiled floor



SPILL confidence 0.93 3.5s

water ·medium size

A tipped-over glass sits on a wood-grain floor, with a clear liquid puddle spreading from its rim. The puddle has distinct raised, reflective edges and darkens the wood beneath it.

Clear water from a tipped glass on wood grain. Opus caught it, Haiku called it a reflection



SPILL confidence 0.60 4.2s

food or drink ·medium size

A distinct brown puddle consistent with spilled coffee lies next to a tipped cup on a white surface, with a hand wiping it with a pink cloth. This is a watermarked stock photo, and the plain white surface could be a table rather than the floor, which lowers confidence.

Tea spill next to a cup. Flagged at low confidence (0.60), which the robot should treat as 'go check'



SPILL confidence 0.90 2.4s

water · small size

A clear, raised puddle with defined meniscus edges sits on a white glossy surface, surrounded by scattered droplets, consistent with spilled water. The surface looks like a close-up floor, so this is likely a small water spill.

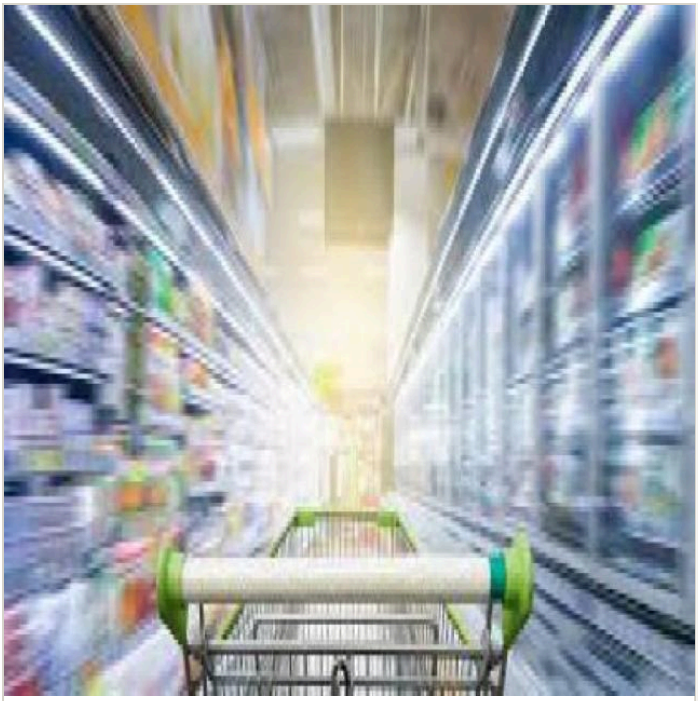
Small clear puddle on a pale floor, almost invisible at a glance



CLEAR confidence 0.88 3.0s

The supermarket aisle floor is partly visible between the shopping carts and shows a uniform grey surface with only light sheen and cart shadows. No puddles, drips or debris are visible, though much of the floor is occluded by carts and people.

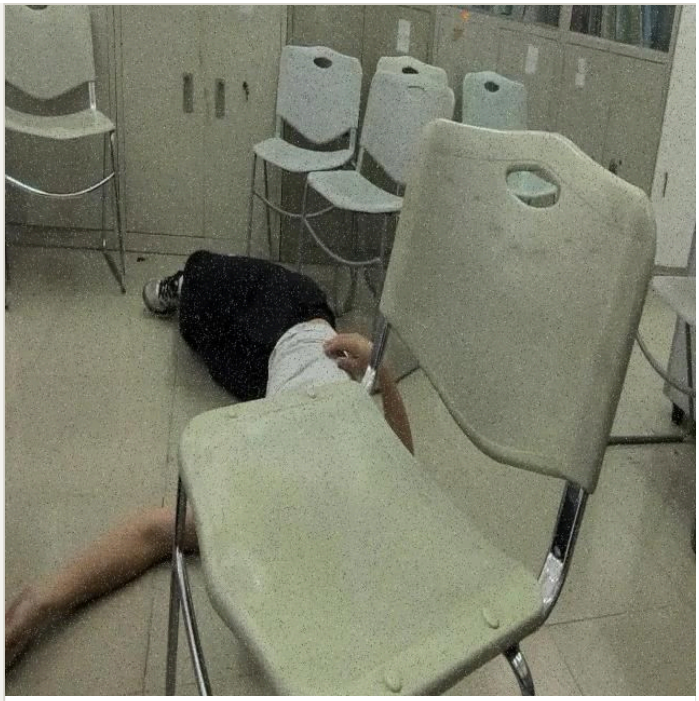
Supermarket CCTV frame, shoppers in an aisle, dry floor



CLEAR confidence 0.90 2.5s

Motion-blurred supermarket freezer aisle with a shopping cart blocking most of the floor; the small visible floor strips show only bright reflections and no distinct puddle or liquid.

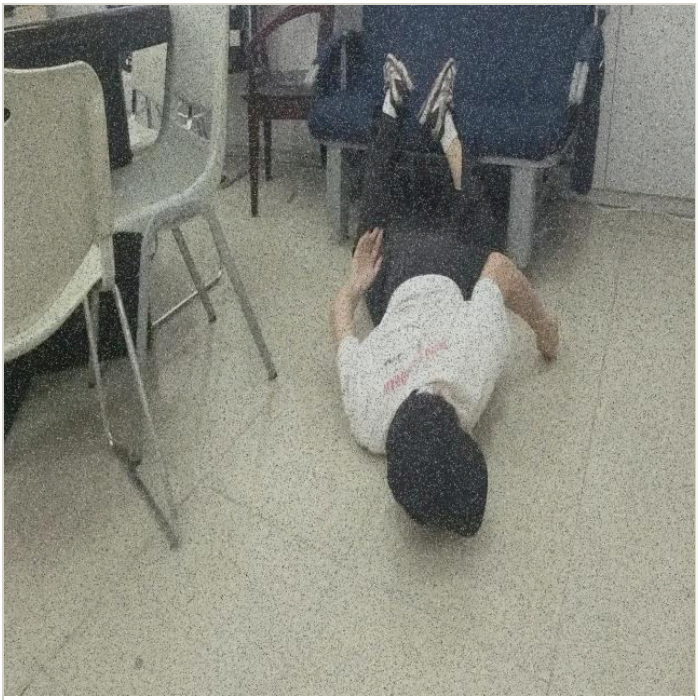
Bright supermarket aisle, polished floor, no spill



CLEAR confidence 0.85 5.1s

No puddle, smear or wet patch is visible on the beige tile floor; only chairs, cabinets and a few small dark specks are present. IMPORTANT: a person appears to be lying motionless on the floor among the chairs, a possible medical emergency that should be escalated to staff immediately.

Person lying on a tiled floor between chairs, no liquid



CLEAR confidence 0.85 2.8s

The light tiled floor shows no puddles, drips or wet outlines, only image noise and soft reflections. A person is lying face-down on the floor with their legs on a sofa, which is not a spill but may be a fall or medical incident and warrants flagging to staff.

Same room, different angle. Scuffed tiles, no spill



CLEAR confidence 0.80 2.6s

Very dark, heavily noisy frame showing a child sitting in a shopping cart; the visible floor area shows no distinct puddle or wet outline, though low light and noise limit certainty.

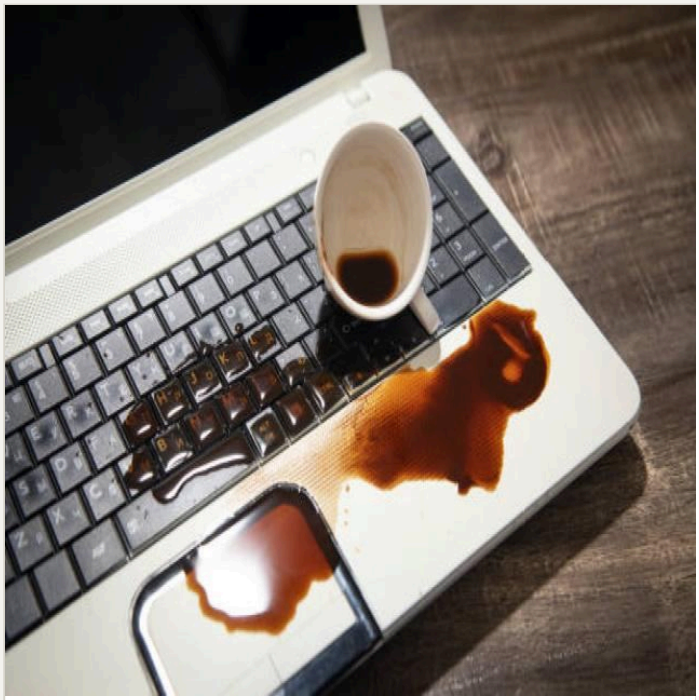
Very dark, grainy frame of a shopping trolley



CLEAR confidence 0.90 2.6s

Outdoor parking-lot sidewalk crowded with shopping carts; the visible concrete and red-brick paving looks dry, with no puddles, drips or wet sheen. Dark patches are cart shadows and mulch, not liquid.

Shopping trolleys on wet-look pavers outdoors



CLEAR confidence 0.90 2.7s

Coffee is spilled on a laptop keyboard and touchpad resting on a wooden table, not on the floor. No floor surface is visible, so this is not a floor spill for the robot to clean.

Coffee on a laptop keyboard. Dataset says 'water', the model says 'not on a floor'

CLOSE CAMERA · MODEL CHOICE

Why Opus 5.5 and not the cheaper model

Claude Haiku 5.5 costs about a thirtieth as much. On the same 200 frames its headline score is close, but it is less consistent: it ignored the floor-only rule on 4 of the 8 mislabelled frames, gave up on several abstract splash images, and missed the one genuinely hard floor spill in the set.



Claude Opus 5.5

SPILL

confidence 0.93

A tipped-over glass sits on a wood-grain floor, with a clear liquid puddle spreading from its rim. The puddle has distinct raised, reflective edges and darkens the wood beneath it.

Claude Haiku 5.5

CLEAR

confidence 0.85

The frame shows a clear glass in the foreground and its reflections and refractions on a glossy, light-colored floor. The distorted shapes on the floor are reflections or shadows of the glass rather than liquid, and no puddle or wet patch is visible.

	CLAUDE OPUS 5.5	CLAUDE HAIKU 5.5
Floor-frame accuracy	100%	97%
Floor spills missed	0 of 95	5 of 95
False alarms	0 of 97	0 of 97
Followed the floor-only rule	8 of 8	4 of 8
Cost, 200 frames	$1.16	$0.032
Mean latency	3.1 s	2.0 s

Recommendation: Opus 5.5 for the proof of concept and for labelling anything subtle. Haiku may later take the easy frames in bulk, with Opus re-checking anything it marks below 0.8 confidence.

CLOSE CAMERA · APPENDIX

Every frame in the evaluation

All 200 frames the model was scored on, spills first, highest confidence first. The 20 frames in the gallery above were picked by hand from this set to show one of each situation; this list is the unfiltered whole. Hover a frame for the model's reason.

Model: claude-opus-5-5, effort low, structured JSON output.

Fixed-camera footage: five mall clips from Pixabay and Pexels (Pixabay Content License, Pexels License), real floors with rendered spills; no AI image generation was used anywhere in this work.

Close-camera data: Spill_Detection v4 by slipp on Roboflow Universe, CC BY 4.0. Validation split, 200 frames, random seed 0.

Report generated from data/results by scripts/build_report.py.